



Introduction to Excel

Describe Excel sheet of
experiments



Outlines:

- **Give general view for Excel sheet.**
- **Useful functions In Excel.**
- **Construction of a Graph in Excel.**
- **Useful Properties in Excel.**

General view of Excel sheet:

- **In physics lab, data collected from experiment setups recorded and processed in certain excel sheets that will upload to the lab google classroom.**
- **How student enter his personal data.**
- **General view for “Tasks”.**
- **General view for “Discussion Questions”**

Useful functions in Excel (**Data processing**):

- The data collected need to be processed and organized as shown.

The average values of the physical parameter represented on x-axis

The uncertainties of the physical parameter represented on x-axis

$\bar{x}$	$\bar{y}$	σ_x	σ_y

The average values of the physical parameter represented on y-axis

The uncertainties of the physical parameter represented on y-axis

Useful functions in Excel (**Data processing**):

○ **Average** (= *AVERAGE*(No1; No2; ...)).

(https://www.youtube.com/watch?v=dZk03_tBT00)

○ **Uncertainty** (= *STDEV.S*(No1; No2; ...)).

(<https://www.youtube.com/watch?v=mseCUOgdwBk>)

○ **Significant Figure** (= *ROUND*(Number; no of digits)).

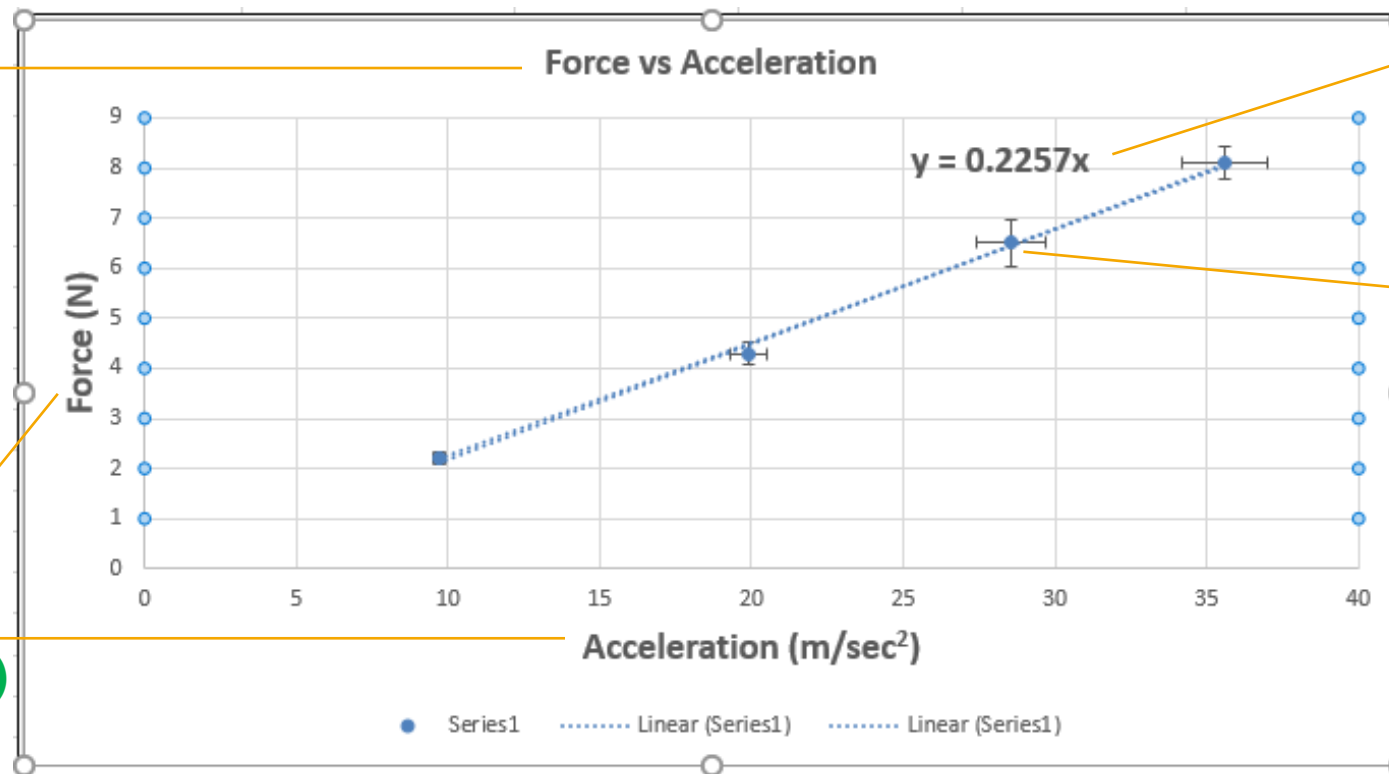
(<https://www.youtube.com/watch?v=q7UdMV2eqA>)

Construction of a Graph in Excel:

- The data collected needed to be represented in a graph as shown.

Eqn describe the curve

Curve title



Points:

- solid circle is average $(\bar{x}_i, \bar{y}_i)$
- Cross is the error bar $((\sigma_x)_i, (\sigma_y)_i)$

Axis titles (parameter, unit)

Construction of a Graph in Excel:

1. **Select graph.**
2. **Enter data.**
3. **Set the shape of graph.**
4. **Set the error bars.**
5. **Enter axis titles.**
6. **Set the intercept.**
7. **Display equation on chart.**
8. **Determine slope.**

(<https://www.youtube.com/watch?v=hwqq0HYrers>)

Useful functions in Excel:

- Repeat a certain formula for multiple cell.

(<https://www.youtube.com/watch?v=4yaceJErMw4>)

- How to write the formula “ $Y \pm x$ ”. (Type = $Y \& \text{“} \pm \text{”} \& x$)